Phase-3 CPU Embeddings — end-to-end proof (§11.4.108 / §11.4.5 / §11.4.69)

Date: 2026-07-06 · Track (`T1/feature/helixllm-full-extension`) · Design: `docs/research/07.2026/00_master/EMBEDDINGS_PROVIDER.md`

Verdict (honest, §11.4.6)

CAPABILITY PROVEN. A real CPU embeddings service booted via the **containers submodule orchestrator** (§11.4.76, rootless podman §11.4.161, NO GPU) served real, deterministic, semantically-ordered embeddings. The §11.4.108 runtime signature is **GREEN-OK**:

```
[RUNTIME-SIGNATURE] PASS  dim=384  |A|=1.0000 |A'|=1.0000 |U|=1.0000
                   cos(A,A')=0.7509   cos(A,U)=0.3931   margin=0.3578  (required ≥ 0.15)
[DETERMINISM]      PASS  3 vectors byte-identical across two identical requests
GREEN-OK
```

(`A` vs `A'` = related/paraphrase pair; `A` vs `U` = unrelated pair. Related similarity » unrelated, margin 0.3578 » the 0.15 floor. Real HTTP 200, non-zero norm, evidence `11_green_proof.txt`, `30_embed_1.txt`, `green_response_{1,2}.json`.)

Lane (honest substitution, §11.4.6 — `23_substitution.txt`)

- **Primary (design default) `nomic-ai/nomic-embed-text-v1.5` FAILED to boot** — TEI `cpu-1.9` rejects its `config.json`: *"duplicate field* `max_position_embeddings`*"* (container exit 1, `22_teilogs_primary.txt`). Known nomic/TEI parser incompatibility.

- **Fell back to `BAAI/bge-small-en-v1.5` (dim 384) — TEI-native, healthy, served the proof.** Both lanes are in the design; the fallback is the ship-now CPU lane. Follow-up: pin a TEI-`cpu-1.9`-parseable nomic revision or upgrade TEI if 768-dim nomic is wanted.

Analyzer is non-bluff (§11.4.107(10) / §11.4.115)

Proven to genuinely discriminate — it does NOT rubber-stamp: - **RED baseline** (`10_red_baseline.txt`): the dim-1536 zero-vector gateway stub → correctly **FAIL** (zero-norm, NaN cosine). Defect reproduced (§11.4.115). - **Golden-BAD fixtures** all correctly **FAIL**: zero-vector, shuffled-order (margin −0.3578), wrong-dim. The analyzer rejects every degraded input. - **Real GREEN input** correctly **PASS** (above).

Full self-validation GREEN (P3-EMB-1 RESOLVED, 2026-07-07)

The complete §11.4.107(10) self-validation now PASSES on the served lane — `40_verify_improved_harness.log`:

```
[SELF-VALIDATION] PASS: analyzer PASSes golden-good and FAILs all
golden-bad fixtures
selfvalidate_exit=0
```

An earlier run FALSE-NEG'd golden-good because the fixture dim was baked to the nomic primary lane (768) while the served fallback lane is bge-small (384). The harness now makes the golden-good lane-dim-aware, AND fixes the boot root cause (removed `WithForceRecreate`, which on this host's podman-compose shim left the pod unstarted → the earlier empty `21_health_primary.txt`; freshness is now a unique per-lane project name + pre-clean boot-down, §11.4.108/§11.4.139) + an exited-only fast-fail poll + a persistent HF cache volume. Boot is now reliable (`health OK after 2 polls`), runtime signature GREEN-OK, self-validation GREEN. P3-EMB-1 CLOSED.

Reproduce

`cd harness && ./run_proof.sh` (boots TEI via the containers submodule, POSTs the sentence triple, asserts the cosine signature + determinism + self-validation, tears the container down single-owner §11.4.119, leaves `helixllm-coder` untouched). `HEALTH_TIMEOUT` env tunes the per-lane health wait. The `harness/phase3embed.bin` is a build artifact (gitignored §11.4.30; rebuild via `go build`).

Composition

§11.4.76 (containers submodule) · §11.4.161 (rootless) · §11.4.108 (runtime signature) · §11.4.107(10) (self-validated analyzer) · §11.4.115 (RED-first) · §11.4.119 (single-owner teardown) · §11.4.6 (honest substitution + honest limitation) · §11.4.147 (conductor completion of the stopped subagent's proof).